	Phase 1	Phase 2
Person 1	DataLoader + Preprocess	LCMA Loss + Augmentation
Person 2	Whisper + LoRA + Content Head	Re-encoder + Full Evaluation
Person 3	Accent Head + Swap + Classifier	Full Training + Ablations
Person 4	Baseline + Loss + Train Skeleton	Results + Plots + Full Report

Phase 1 — What is Parallel vs Sequential

```
PHASE 1
──────────────────────────────────────────────────────────────

Person 1  ═══════════════════  DataLoader + Preprocess
          (fully independent, start immediately)

Person 2  ═══════════════════  Whisper + LoRA + Content Head
          (fully independent, start immediately)

Person 3  ═══════════════════  Accent Head + Swap + Classifier
          (fully independent, start immediately)

Person 4  ═══════════════════  Baseline WER + Loss + Train Skel
          (fully independent, start immediately)

          ←──── ALL 4 WORK IN PARALLEL ────→
          No one waits for anyone in Phase 1

          ▼ CONNECT POINT (1 session together)
          All 4 plug pieces together and run one forward pass
          Fix shape errors together
```

Phase 2 — What is Parallel vs Sequential

```
PHASE 2
──────────────────────────────────────────────────────────────

Person 1  ═══════════  LCMA Loss
          Independent — just needs Content Head output shape
          (Person 3 tells Person 1 the shape → done)

Person 2  ═══════════  Re-encoder
          Needs Person 3's Swap output to feed back
          WAIT for Accent Swap to be ready first

Person 3  ═══════════════════════════  Training + Ablations
          Needs Person 1 LCMA Loss ready
          Needs Person 2 Re-encoder ready
          THEN can start training
          ← this is the bottleneck

Person 4  ═══════════  Results + Plots + Report
          Needs Person 3 training to finish
          THEN evaluate and write report

TIMELINE:
Week 1 of Phase 2:   Person 1 + Person 2 work in parallel
                     Person 3 + Person 4 prepare training code
Week 2 of Phase 2:   Person 3 starts training (overnight runs)
                     Person 4 starts report writing in parallel
Week 3 of Phase 2:   Person 3 finishes training
                     Person 4 fills in results and finalizes report
```

PHASE 1 TASKS

Task 1 — DataLoader + Preprocess

What:
Download datasets, convert audio into numbers the model can read, and organize into batches.

Tasks

- Download **L2-ARCTIC** and **Common Voice** from HuggingFace
- Filter by accent:
  - Indian
  - Mandarin
  - Korean
  - Arabic
- Load WAV files and **resample to 16 kHz**
- Convert audio to **Log-Mel Spectrogram [T × 80]**
- Normalize spectrogram
  - Mean = 0
  - Standard deviation = 1
- Split dataset
  - 80% Train
  - 10% Validation
  - 10% Test
- Build **PyTorch DataLoader** returning:
  - spectrogram
  - transcript
  - accent_id

Goal
Print one batch and confirm:

- Spectrogram shape = **[32 × 80 × T]**
- Transcripts are readable strings
- Accent IDs are numeric

Parallelization:
Fully independent — can start immediately.

Task 2 — Whisper + LoRA + Content Head

What:
Load the pretrained Whisper model, freeze its weights, add LoRA adapters, and attach a content projection head.

Tasks

- Load **Whisper-small** from HuggingFace

- Freeze all Whisper weights
- Inject **LoRA adapters** into encoder attention layers
    - r = 16
    - lora_alpha = 32
- Verify trainable parameters are **< 1% of total**
- Run a forward pass
    - Confirm encoder output shape **[B × T × 768]**
- Build **Content Projection Head**
    - Linear(768 → 256)
    - LayerNorm
    - ReLU
- Confirm output shape **[B × T × 256]**

Goal

- Whisper loads successfully
- LoRA adapters injected
- < 1% parameters trainable
- Encoder output and content vector shapes correct

Parallelization:
Fully independent — can start immediately.

Task 3 — Accent Head + Swap + Classifier

What:
Build the three networks responsible for extracting and manipulating accent information.

Tasks

- Build **Accent Projection Head**
    - Linear(768 → 128)
    - LayerNorm
    - ReLU
    - MeanPool across time dimension
- Confirm accent vector shape **[B × 128]**
- Build **Accent Swap Function**
    - Combine content vector with a neutral accent vector
- Confirm swapped output shape **[B × T × 384]**
- Build **Accent Classifier Head**
    - Linear(128 → 64)
    - ReLU
    - Linear(64 → 4)
    - Softmax
- Classifier predicts accents:
    - Indian
    - Mandarin
    - Korean
    - Arabic

Goal

- All modules run without errors
- Tensor shapes correct
- Classifier outputs **4 probabilities summing to 1**

Parallelization:
Fully independent — can start immediately.

Task 4 — Baseline + Loss + Training Skeleton

What:
Establish the baseline performance and build the training framework.

Tasks

- Run **Whisper-small zero-shot** on **L2-ARCTIC**
- Compute **WER per accent**
    - Hindi
    - Mandarin
    - Korean
    - Arabic
- Save results to **baseline_wer.json**

Build training components:

- Implement **combined loss**

Total Loss =
ASR Loss + α × Accent Loss + β × LCMA Loss

- Implement **training loop skeleton**
    - Epoch loop
    - Batch loop
    - Forward pass
    - Compute loss
    - Backpropagation
    - Weight update
    - Loss logging
- Write **evaluation script**
    - Load checkpoint
    - Run on test set
    - Print WER per accent

Goal

- Baseline WER recorded
- Training skeleton runs **one batch without crashing**
- Loss prints to screen

Parallelization:
Fully independent — can start immediately.

PHASE 2 TASKS

Task 5 — LCMA Loss + Augmentation

What:
Implement the cycle-consistency penalty and add training augmentations.

Tasks

- Implement **LCMA Loss**
  - MSE between original content vector and re-encoded content vector
- Interpretation
  - High loss → accent leaked into content
  - Low loss → clean separation
- Tune **β (beta)** weight controlling LCMA Loss
- Add **Pitch Shift Augmentation**
- Add **Speed Perturbation**
  - Random speed between **0.9× and 1.1×**
- Add **SpecAugment**
  - Randomly mask frequency bands
  - Randomly mask time steps
- Handle class imbalance
  - Oversample smaller accent groups

Goal

- LCMA Loss decreases during training
- Augmentation applied **only during training**

Parallelization:
Fully independent in Phase 2.

Task 6 — Re-encoder + Full Evaluation

What:
Complete the cycle consistency loop and run final evaluations.

Tasks

- Build **Re-encoder**
  - Pass swapped features through Whisper encoder
- Extract re-encoded content vector
  - Using same Content Projection Head
- Feed result into **LCMA Loss**
- After training:
  - Run evaluation on **L2-ARCTIC test set**
  - Compute **WER per accent**
- Run evaluation on **EdAcc dataset**

    - Tests unseen accents
- Send evaluation results to Task 8

Goal

- Re-encoder output matches original content vector shape
- WER computed for all accents on both datasets

Parallelization:
Waits 2–3 days for Accent Swap implementation.

Task 7 — Full Training + Ablations

What:
Train the full system and run ablation experiments.

Tasks

- Connect full pipeline

DataLoader → Encoder → Content Head → Accent Head → Swap → Re-encoder → LCMA Loss → ASR Loss → Combined Loss

Training setup

- Optimizer: **AdamW**
- Learning rate: **1e-4**
- Train **3–5 epochs** on Common Voice

Training monitoring

- Save checkpoint each epoch
- Monitor validation WER
- Apply early stopping

Ablation experiments

1. Train **without Accent Swap**
2. Train **without LCMA Loss**
3. Train **full MirrorSpeech model**

Record **WER per accent** for all runs.

Goal

- Training loss decreases each epoch
- Full model **outperforms both ablations**

Parallelization:
Starts after Task 5 and Task 6 are ready.

Task 8 — Results + Plots + Final Report

What:
Compile results, generate figures, and prepare the final submission.

Tasks

- Collect WER results from Tasks 6 and 7
- Build **results table**
  - Baseline Whisper vs MirrorSpeech
  - Per-accent improvement
- Plot:
  - Training loss curve
  - WER bar chart (before vs after)
  - Ablation comparison chart
- Write **final project report**
  - Introduction
  - Related Work
  - Methodology
  - Experiments
  - Results
  - Conclusion
- Update presentation slides
- Coordinate final presentation rehearsal

Goal

- Clear WER improvements across all accents
- Clean and readable plots
- Final report and presentation completed

Parallelization:
Starts after training results are available.